TUĞÇE MÜLLER

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Research interest: generative AI in adult and lifelong learning

EDUCATION

M.A. Adult Education and Management in Lifelong Education from 04/2027
Julius-Maximilians-Universität Würzburg, Germany · Medium of instruction: English

B.Sc. Computer Education and Instructional Technology 09/2020 – 06/2026
Middle East Technical University (METU), Faculty of Education, Ankara, Türkiye · Medium of instruction: English

- Including a one-year English preparatory programme (2020–2021); undergraduate studies 10/2021 – 06/2026
- Cumulative GPA: 3.73 / 4.00 (1.4 in the German grading system); ranked 3rd of 66 final-year students
- Relevant coursework: instructional design, educational psychology, teaching methods, distance education, measurement and evaluation, research methods, educational statistics (graduate level), data science for education, AI applications in education

Erasmus+ Exchange Semester 02/2024 – 07/2024
Università degli Studi di Salerno, Italy

- Coursework in strategic management, organizational change, decision analysis and financial markets

RESEARCH EXPERIENCE

Undergraduate Researcher (Artificial Intelligence) 11/2024 – 09/2025
TÜBİTAK 3501 project "Implementation of Large Language Models in Dialogue-Based Peer Feedback and Investigation of Its Effects", METU, Ankara

- Reviewed 500+ research articles on AI in education, feedback systems and programming education
- Supported experimental design, data collection, frequency analysis and research documentation

TEACHING EXPERIENCE

Information Technologies Teacher – Intern 10/2025 – 06/2026
Ministry of National Education, Republic of Türkiye – 23 Nisan Secondary School, Ankara

- Completed 144 hours of teaching practice across two semesters
- Prepared curriculum-aligned lesson plans, activities, worksheets and assessment tasks; taught interactive lessons using active learning and questioning strategies

Undergraduate Assistant 10/2024 – 06/2025
METU, Department of Computer Education and Instructional Technology

- Supported faculty with instructional activities; guided students in designing technology-based learning materials and projects

Volunteer Teaching Assistant 02/2023 – 07/2023
METU – CEIT216 Principles and Methods of Instruction (supervisor: Prof. Dr. İbrahim Soner Yıldırım)

PROFESSIONAL EXPERIENCE

Student Staff – Digital Content and Administrative Documentation (part-time) 08/2025 – 06/2026
METU Personnel Department, Ankara

- Developed and structured the department's official website (pdb.metu.edu.tr) in Drupal, including sections for academic, administrative and international staff, students, forms and announcements
- Prepared and published the digital Academic International Handbook
- Documented the workflows of 9 METU administrative departments with department heads and converted them into standardized process cards for risk management
- Assisted in preparing the department's 2025 annual report using Oracle-based data tables; collaborated with the IT unit on website maintenance

Robotics Intern 07/2023 – 08/2023
ROMER – METU Robotics and AI Research Center, Ankara

- Worked on robotic solutions with ROS; built a Tkinter interface for drone simulation (Python, NumPy, OpenCV)

SELECTED PROJECTS

RoadmapGeneratorAI – LLM-based personalized learning roadmap generator CEIT490
Generates weekly study plans from uploaded course materials · LangChain, Hugging Face Transformers, ChromaDB, Streamlit · github.com/tugceunlu/RoadmapGeneratorAI

NextStepCEIT – student-centred educational guidance website CEIT435–436
Course overviews, learning roadmaps and resources for CEIT students; planned, evaluated and improved with student feedback · nextstepceit.com

Moodle distance-education course and Articulate Storyline teacher-training course CEIT321, CEIT225
15-week online Occupational Health and Safety course; interactive course built with learner and needs analysis, formative evaluation and usability testing

HONORS, CERTIFICATES AND VOLUNTEERING

Honors	METU High Honor (2023–24 Fall, 4.00 GPA; 2021–22 Spring) · METU Honor (2022–23 Fall; 2021–22 Fall)
Certificates	PCEP – Certified Entry-Level Python Programmer, OpenEDG Python Institute (2025) · Python Essentials 1, Cisco (2025) · Foundations of Project Management (2024) and Project Initiation (2024), Google · Youthpass: Mobility of Youth Workers, Erasmus+ (France 2022, Greece 2022)
Volunteering	LÖSEV – Foundation for Children with Leukemia, volunteer (10/2023 – present)

SKILLS

Languages	Turkish (native) · English (advanced) · German (A2, B1 in progress) · Italian (A1)
Research	Systematic literature review, survey design, data collection, statistical analysis, research reporting
Education	Instructional design, lesson planning, classroom management, assessment, learner and needs analysis, distance-education design
Technical	Python, LangChain, Hugging Face Transformers, ChromaDB, Streamlit, Moodle, Articulate Storyline, Drupal, Oracle-based reporting